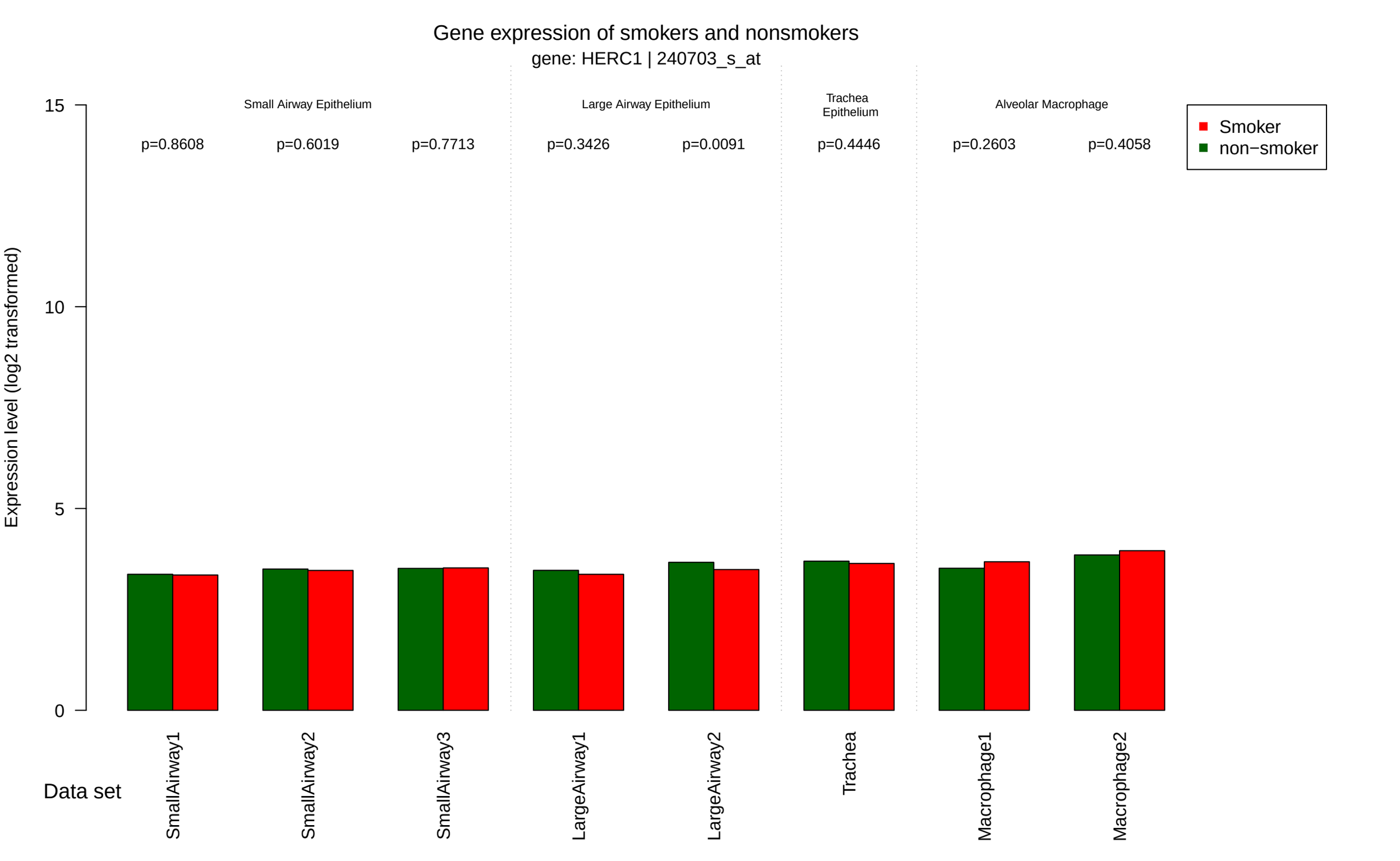
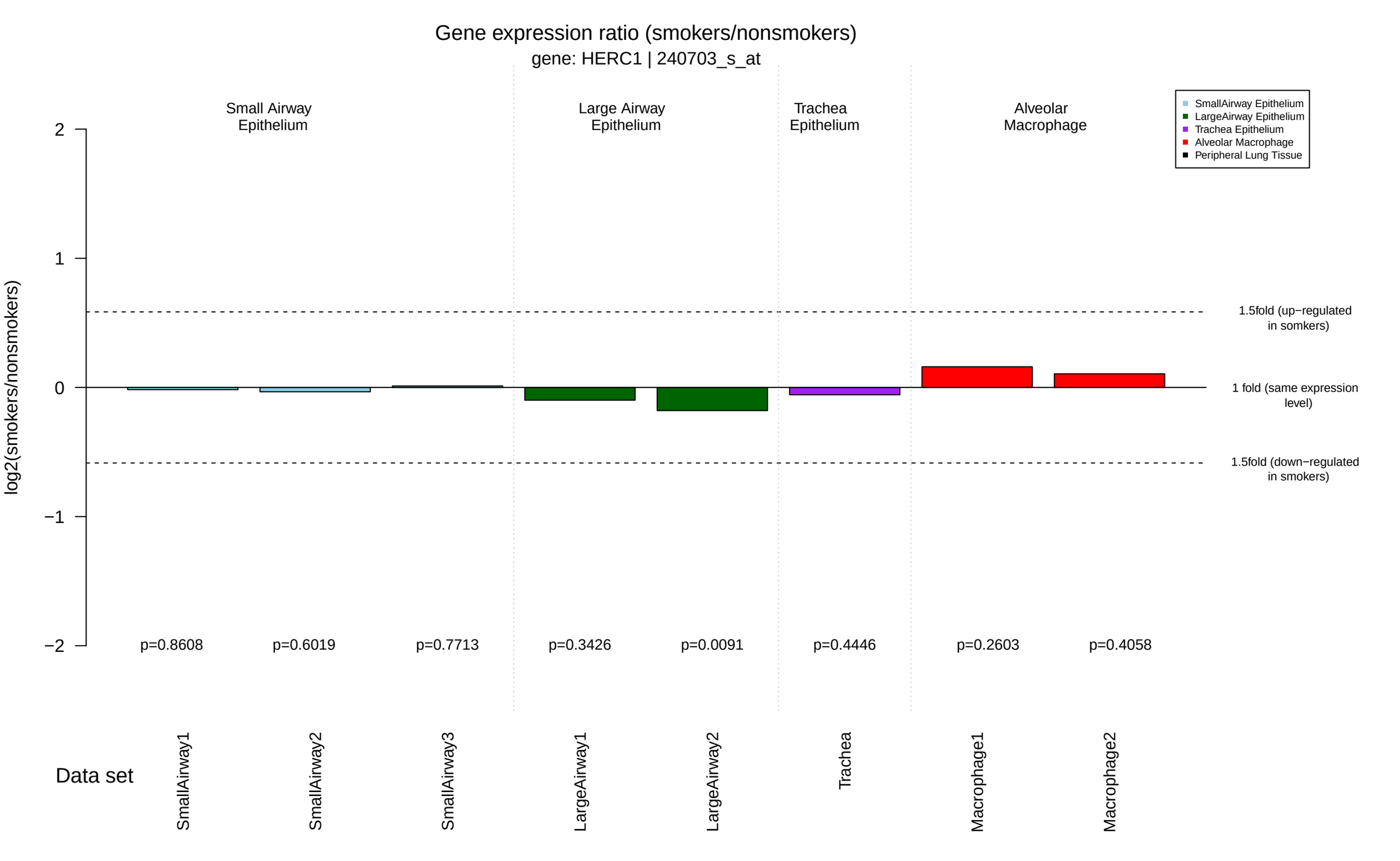
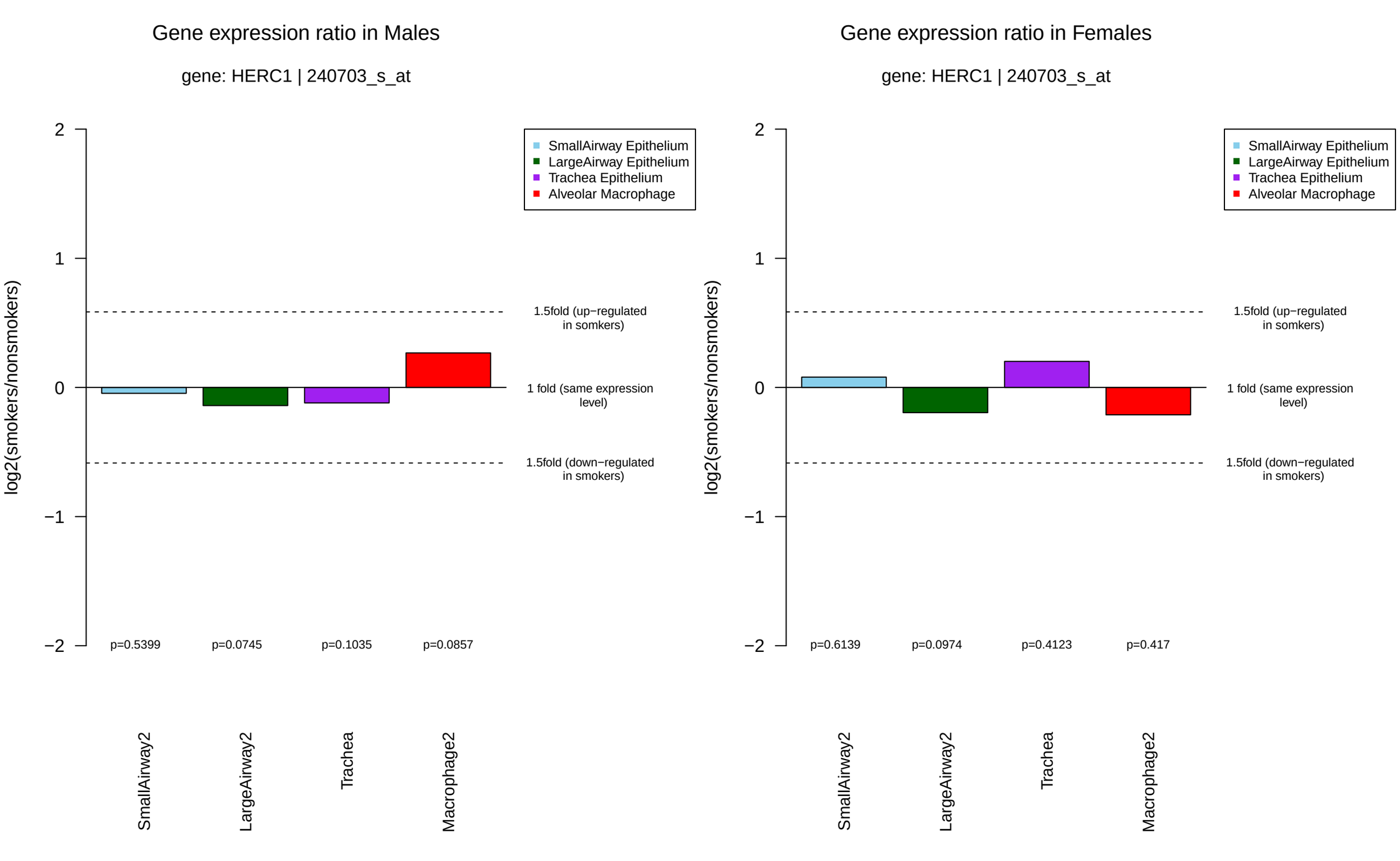
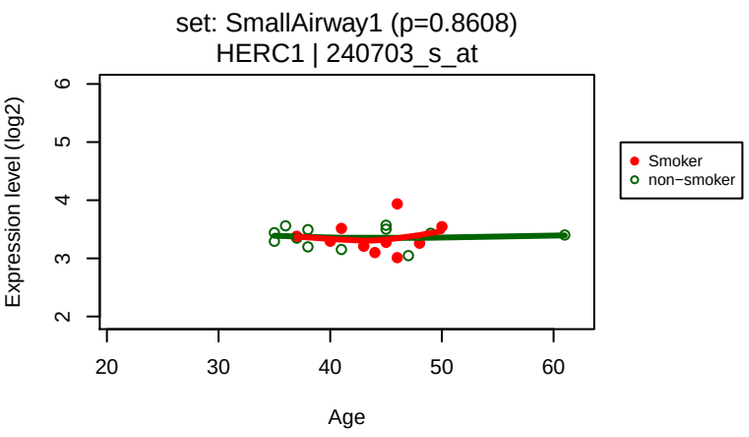
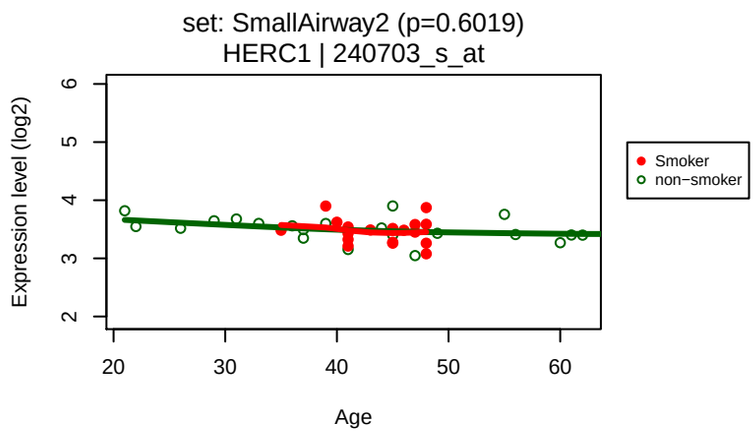
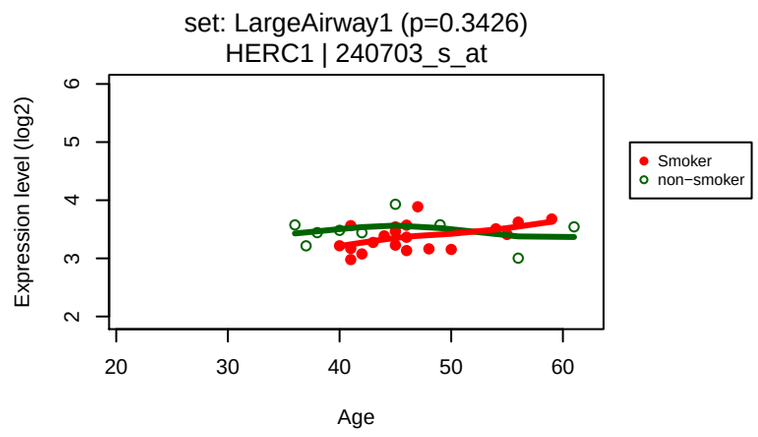

Gene expression of smokers and nonsmokers
gene: HERC1 | 240703_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8608
p=0.6019
p=0.7713
p=0.3426
p=0.0091
p=0.4446
p=0.2603
p=0.4058
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

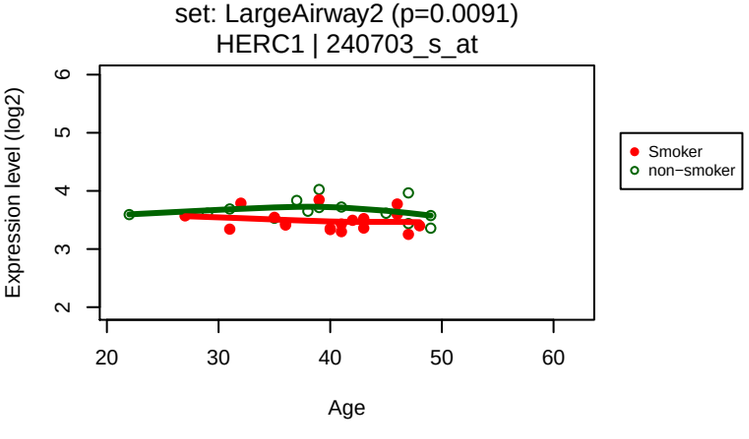

Gene expression ratio (smokers/nonsmokers)
gene: HERC1 | 240703_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8608
p=0.6019
p=0.7713
p=0.3426
p=0.0091
p=0.4446
p=0.2603
p=0.4058
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

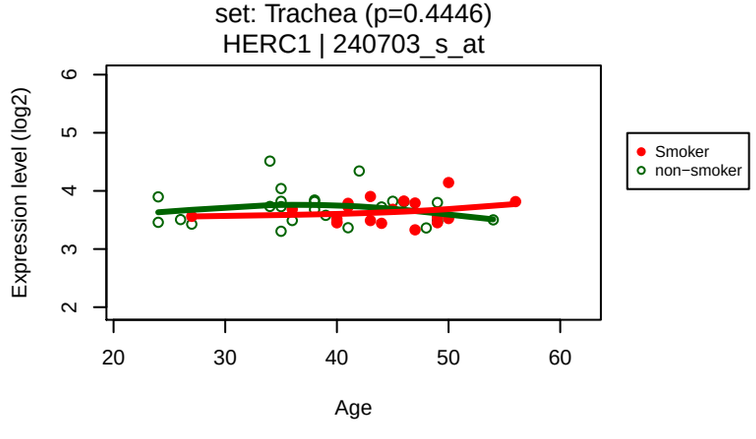

Gene expression ratio in Males
gene: HERC1 | 240703_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5399
p=0.0745
p=0.1035
p=0.0857
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HERC1 | 240703_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6139
p=0.0974
p=0.4123
p=0.417
SmallAirway2
LargeAirway2
Trachea
Macrophage2

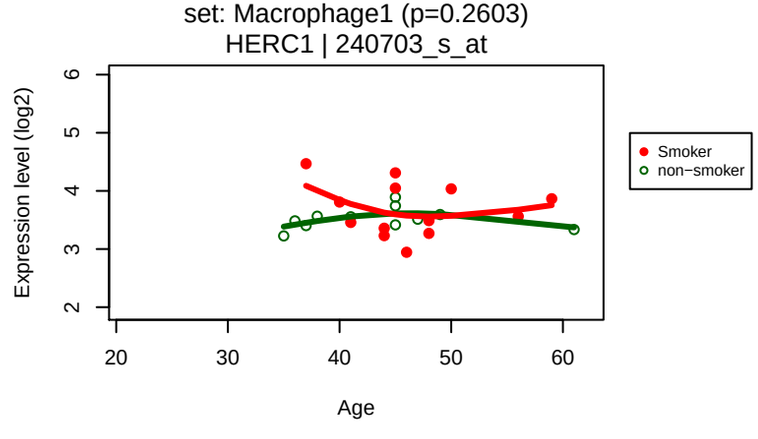

set: SmallAirway1 (p=0.8608)
HERC1 | 240703_s_at
Expression level (log2)
Age
Smoker
non−smoker

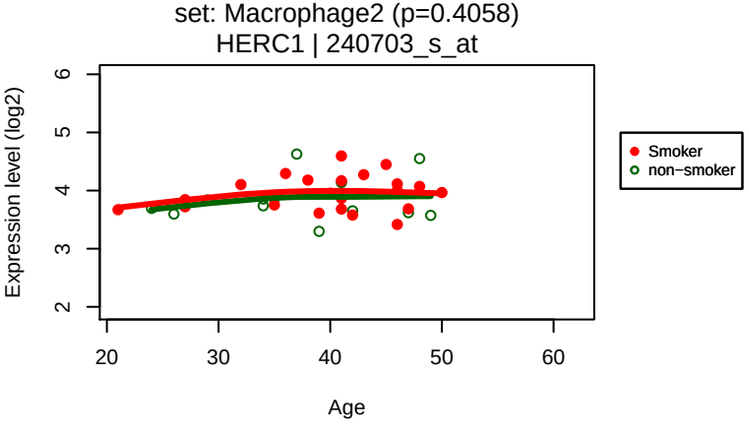

set: SmallAirway2 (p=0.6019)
HERC1 | 240703_s_at
Expression level (log2)
Age
Smoker
non−smoker

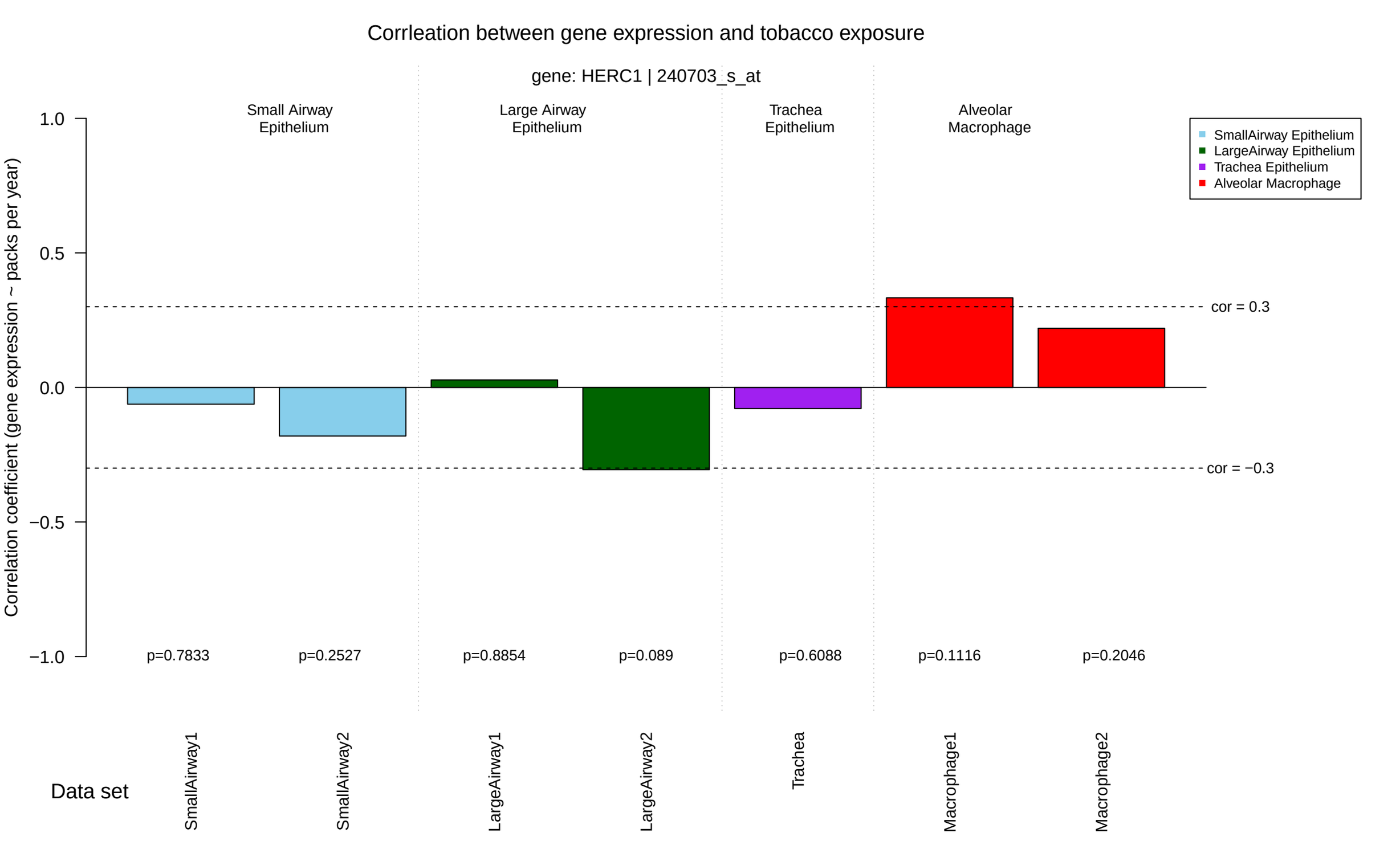

set: LargeAirway1 (p=0.3426)
HERC1 | 240703_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0091)
HERC1 | 240703_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4446)
HERC1 | 240703_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2603)
HERC1 | 240703_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4058)
HERC1 | 240703_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HERC1 | 240703_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7833
p=0.2527
p=0.8854
p=0.089
p=0.6088
p=0.1116
p=0.2046
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2